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**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 3

Title: RETAIL SALES & CUSTOMER ANALYTICS DASHBOARD

Submitted By:

Name: Keerthi N

Reg. No.:2411021240024

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Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application

Alliance University

Chandapura - Anekal Main Road, Anekal

Bengaluru – 562106

RETAIL SALES & CUSTOMER ANALYTICS DASHBOARD

GitHub-Repo-Link: <https://github.com/keerthi19hub/Business-Analytics>

Objective and Problem Statement

The main objective of this experiment was to connect Power BI with a MySQL retail database and analyse the business data. The data was cleaned and transformed using Power Query, relationships were created between the tables, and DAX measures were used for calculations. Finally, a three-page interactive dashboard was created to understand sales, customers, stores, payment methods, and demographics.

The retail data contains sales transactions along with product, store, and customer details. The dashboard provides management with a simple and centralized view of business performance and customer behaviour.

MySQL Database Creation and SQL Operations

The source data was first prepared in **MySQL Workbench**. A database named **RetailDB** was created and selected using CREATE DATABASE and USE commands. The required tables such as Products, Stores, and Sales were created with suitable columns and data types.

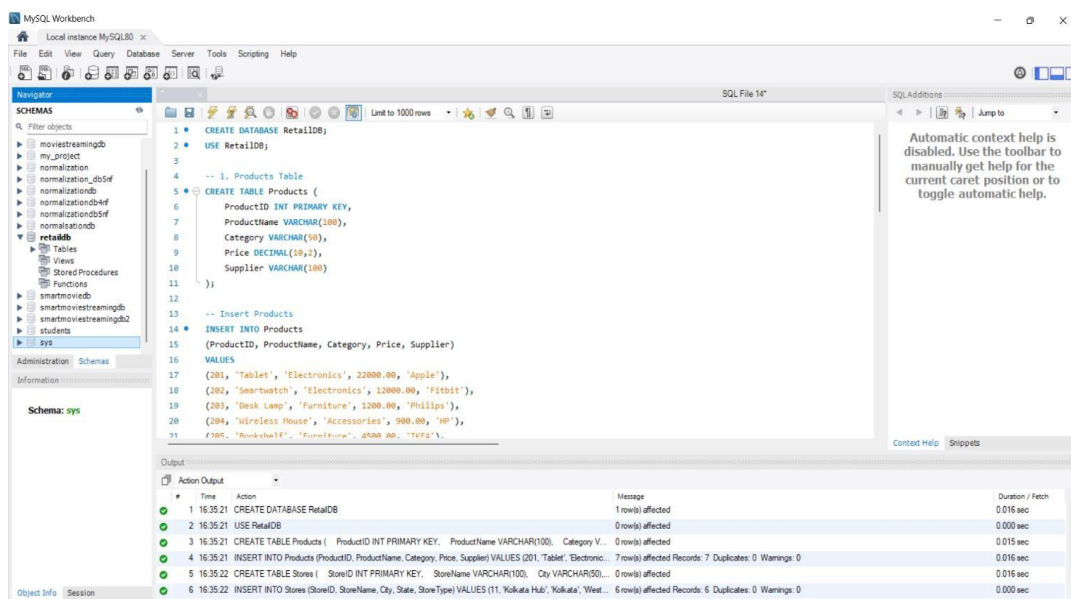
The **Sales table** was used as the main transaction table. It contains details such as SaleID, ProductID, StoreID, Quantity, SaleDate, and TotalAmount. The Products and Stores tables provide additional information related to the sales transactions.

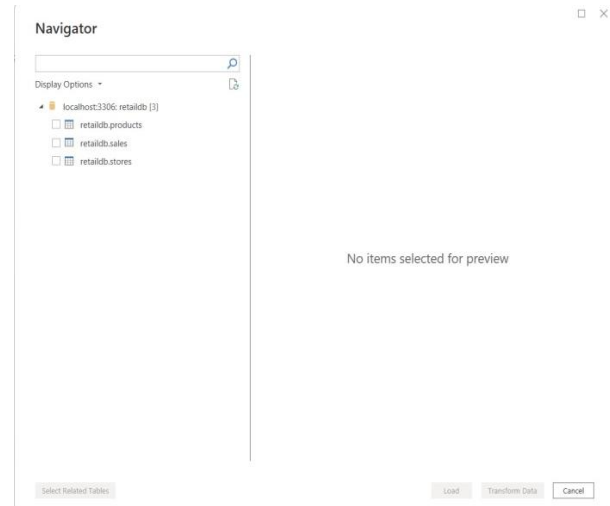
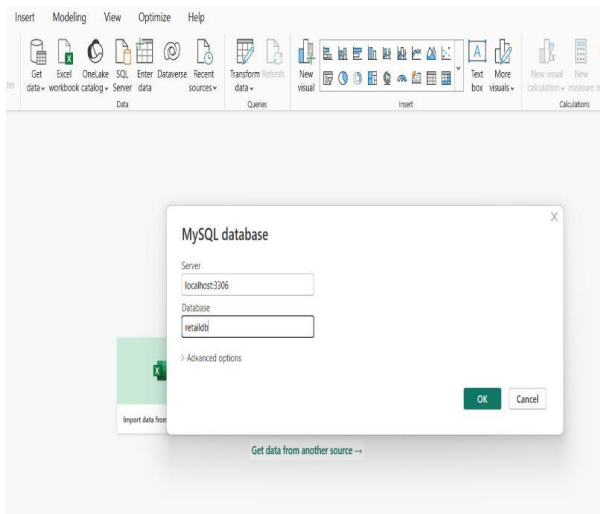
Connecting MySQL to Power BI

Power BI Desktop was connected to the MySQL **RetailDB** database using the **Get Data** option. The required tables were selected and loaded into Power Query for further preparation.

The following tables were imported:

- Sales
- Products
- Stores
- Customer Details
- Calendar





Data Cleaning and Transformation

After importing the data, I used **Power Query Editor** to clean and prepare the datasets. This was done to make the data consistent and suitable for analysis.

Some of the main cleaning and transformation work included:

- Removed duplicate records.
- Checked and handled missing values.
- Renamed columns where required.
- Changed columns to the correct data types.
- Converted SaleDate into Date format.
- Converted TotalAmount and Price into Decimal format.
- Created Age Groups such as **18–25, 26–35, and 36–45**.
- Created Calendar fields such as **Month Name, Month Number, Month Year, and Year**.

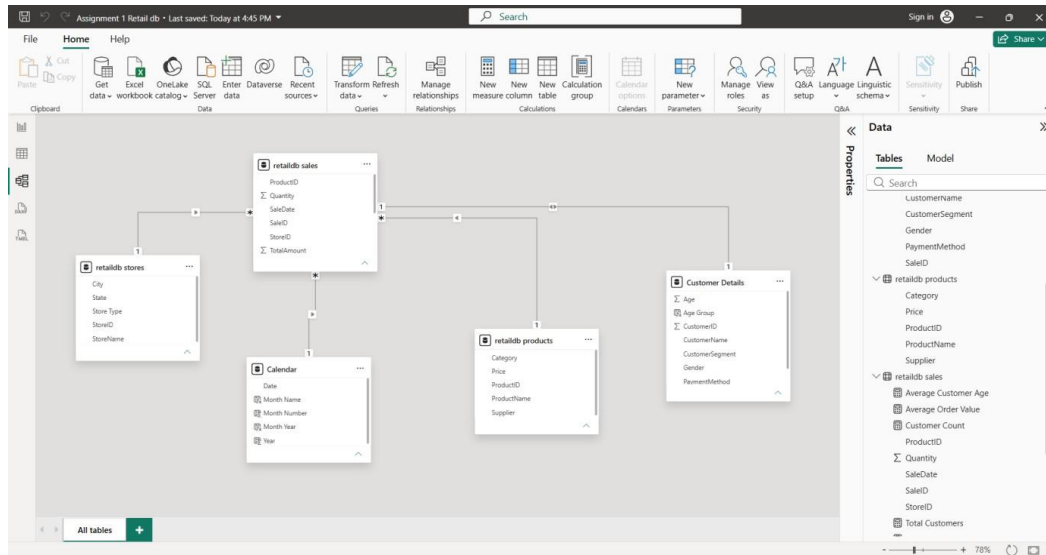
Data Modeling

A **star-schema style data model** was created in Power BI. The Sales table was used as the main fact table, while Products, Stores, Customer Details, and Calendar were used as supporting tables.

The main relationships created were:

- Sales → Products using **ProductID**
- Sales → Stores using **StoreID**
- Sales → Customer Details using **SaleID**
- Sales → Calendar using the **Date field**

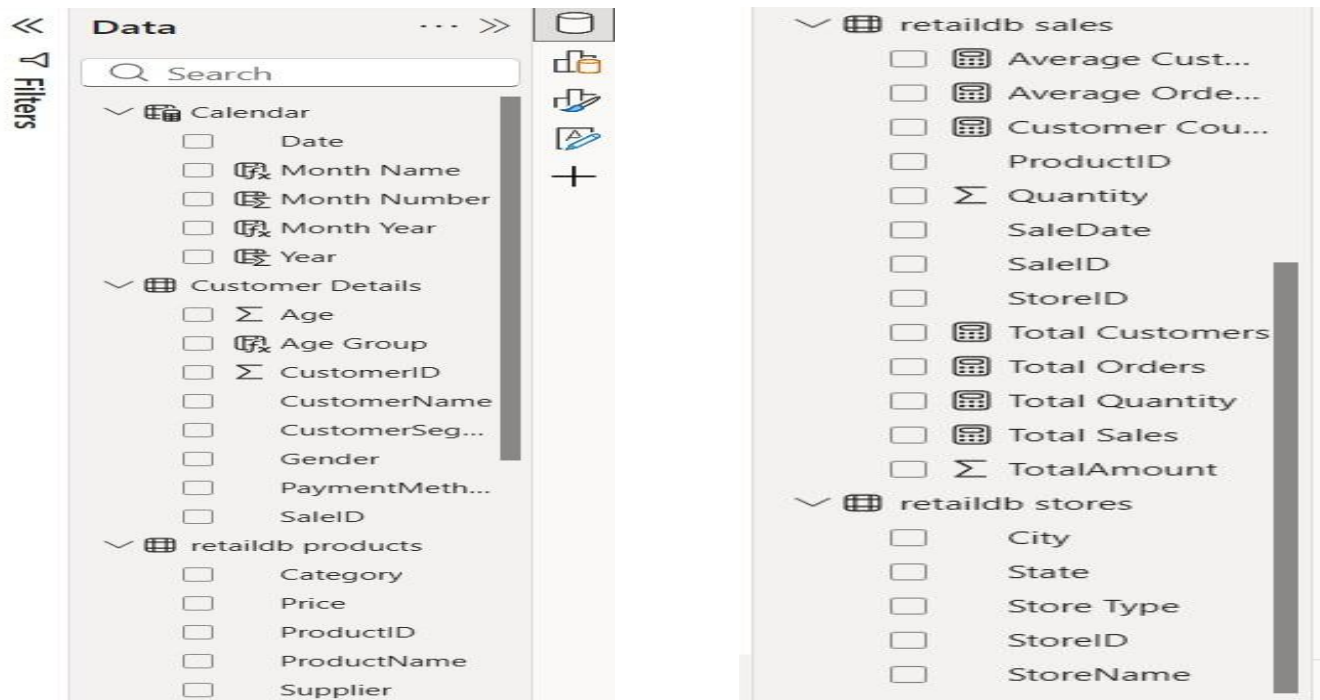
The relationships and filtering directions were checked to make sure the data worked correctly in the dashboard.



DAX Measures Created

I created different DAX measures to perform calculations and display important business information. These measures also make the dashboard interactive when filters are applied.

- **Total Sales** – Calculates the total revenue generated.
- **Total Orders** – Counts the unique sales transactions.
- **Total Quantity** – Calculates the total number of products sold.
- **Total Customers** – Counts the unique customers.
- **Average Order Value** – Calculates the average amount spent per order.
- **Average Customer Age** – Calculates the average age of customers.
- **Customer Count** – Counts the customer records.



Dashboard Development and Explanation

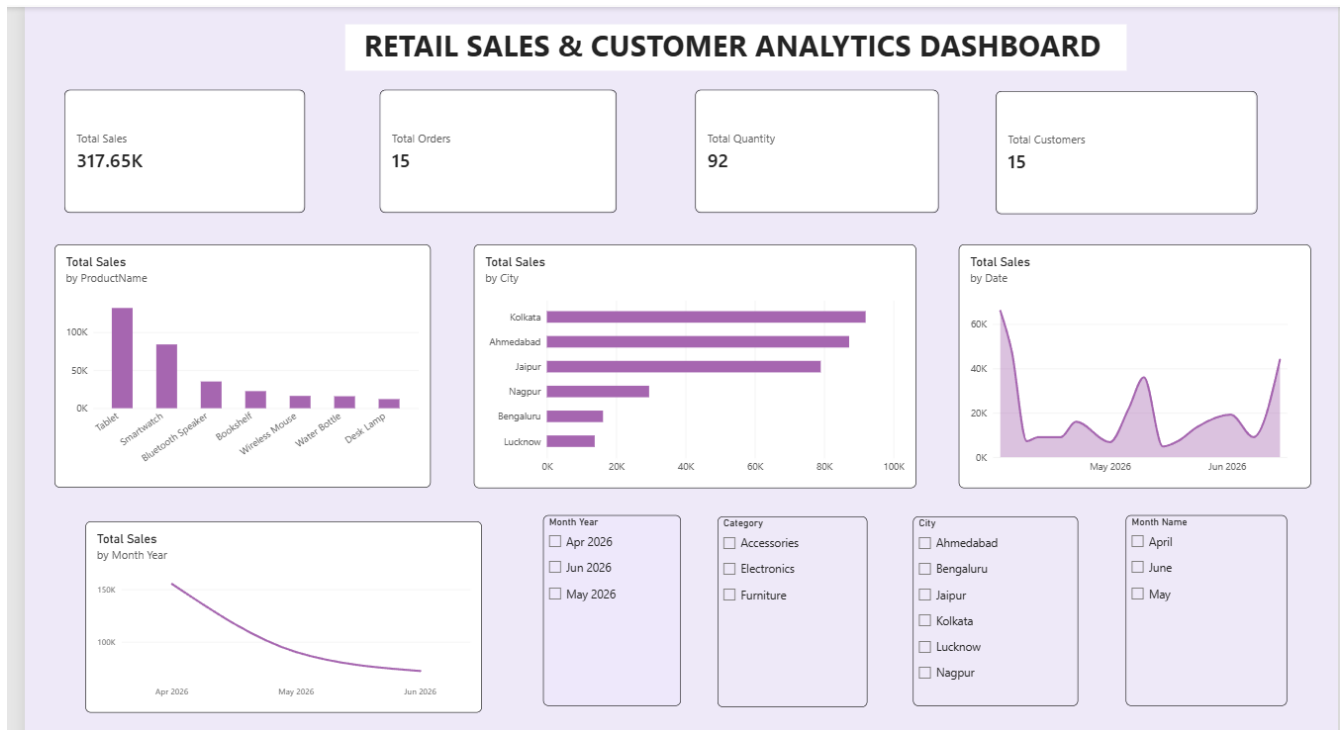
Page 1 – Retail Sales & Customer Overview

The first page is the main overview dashboard. It shows the overall sales and customer performance using KPI cards and different charts.

The main KPI values shown are:

- Total Sales – **317.65K**
- Total Orders – **15**
- Total Quantity – **92**
- Total Customers – **15**

It also includes charts for **product-wise sales**, **city-wise sales**, **sales by date**, and **sales by month-year**. Slicers such as Month Year, Category, City, and Month Name were added to make the dashboard interactive.



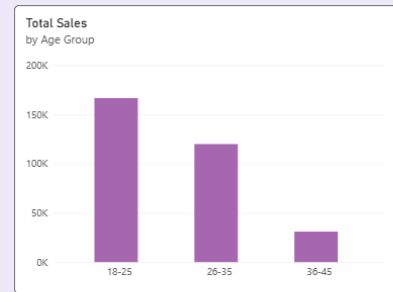
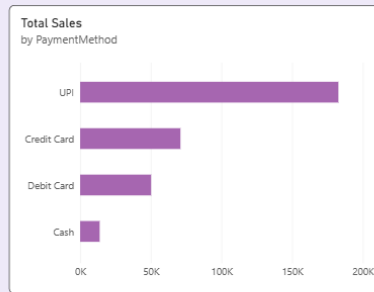
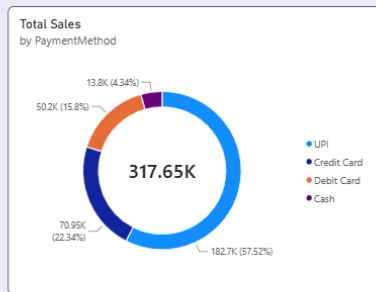
Page 2 – Customer & Store Segmentation

The second page focuses mainly on **customers and stores**. It helps to compare sales based on gender, store type, and customer segment.

The page contains:

- Donut chart for **Sales by Gender**
- Pie chart for **Orders by Store Type**
- Column chart for **Sales by Customer Segment**

The customer segments include **Regular, New, and Premium** customers.



Gender

☐ Female

☐ Male

CustomerSegment

☐ New

☐ Premium

☐ Regular

PaymentMethod

☐ Cash

☐ Credit Card

☐ Debit Card

☐ UPI

Conclusion

Overall, I successfully connected **MySQL with Power BI**, imported and cleaned the retail data, created relationships between the tables, and developed DAX measures for analysis. The final three-page dashboard provides a clear view of **sales, customer behaviour, store performance, payment methods, and demographics**.

The interactive dashboard makes it easier to understand the business data and compare different aspects of retail performance.